

Exploratory Data Analysis: Apple Inc. (AAPL) Stock Price History

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Introduction

This report presents an exploratory data analysis of Apple Inc. (AAPL) stock price history from January 2015 through December 2025, using daily trading data sourced from Yahoo Finance via the tidyquant package in R.

Research Questions

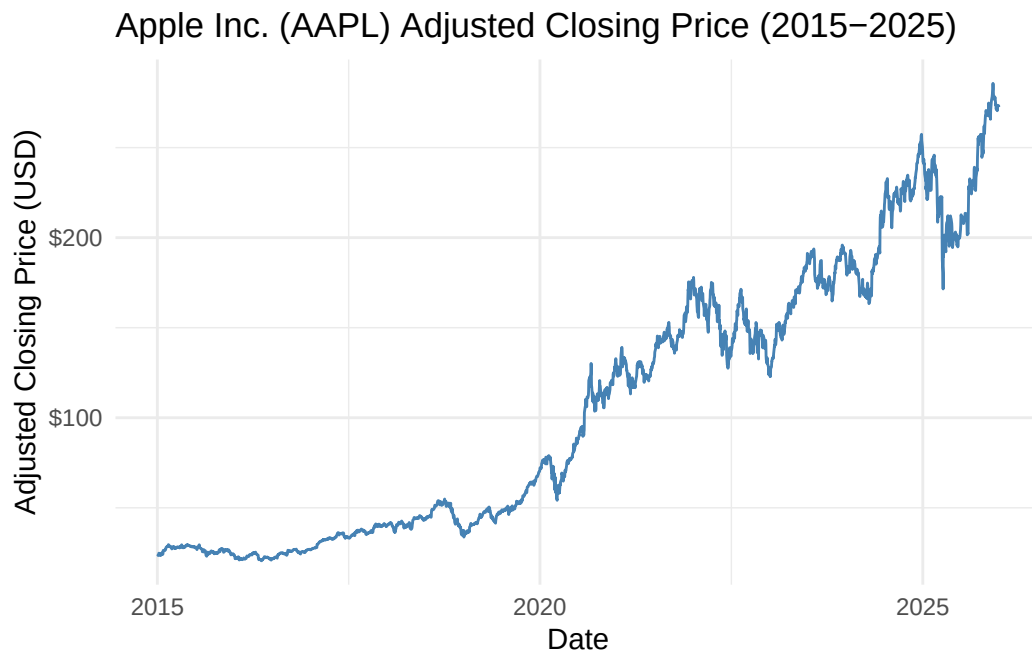
1. How has Apple's stock price trended over the past decade, and what were the major growth periods?
2. How did Apple's annual returns vary year to year, and which years were strongest and weakest?
3. Is there a seasonal pattern in Apple's stock performance across months of the year?
4. How has trading volume changed over time, and when was investor activity highest?
5. How has Apple's stock volatility changed over time, and what events drove peak volatility?

Data Overview

The dataset contains 2,765 daily observations spanning 2015-01-02 to 2025-12-30. Each row represents one trading day and includes eight variables: symbol, date, open, high, low, close, volume, and adjusted close price. There are no missing values in any column. The adjusted close price which accounts for stock splits and dividends is used as the primary measure of price throughout this report. Over the period, Apple's adjusted closing price ranged from a minimum of \$20.57 to a maximum of \$285.66, with a median of \$88.30.

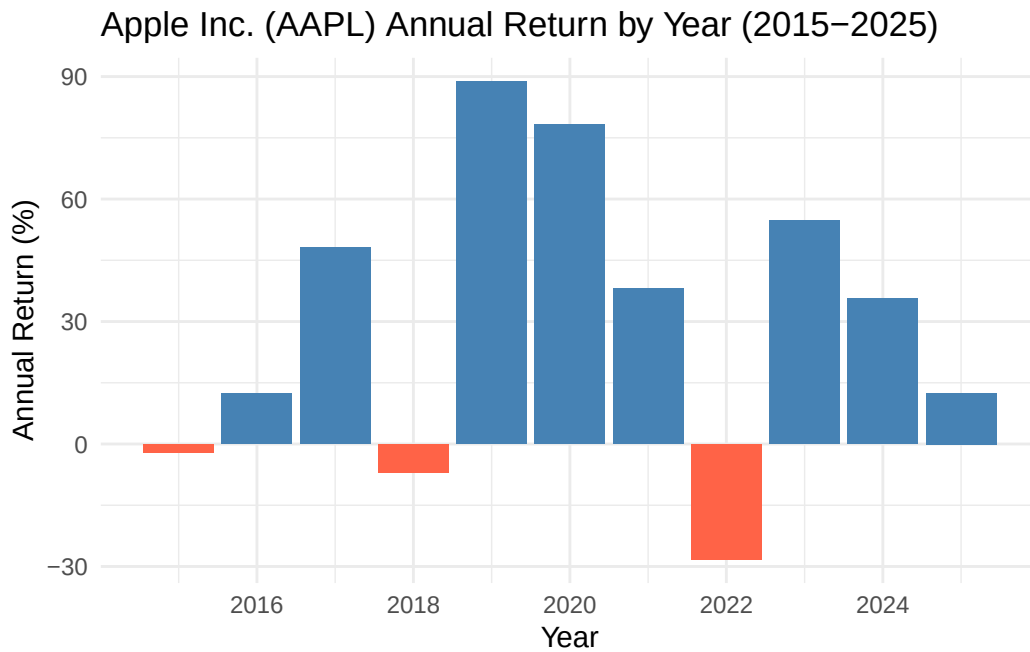
Visualizations and Analysis

Chart 1: Long - Term Price Trend



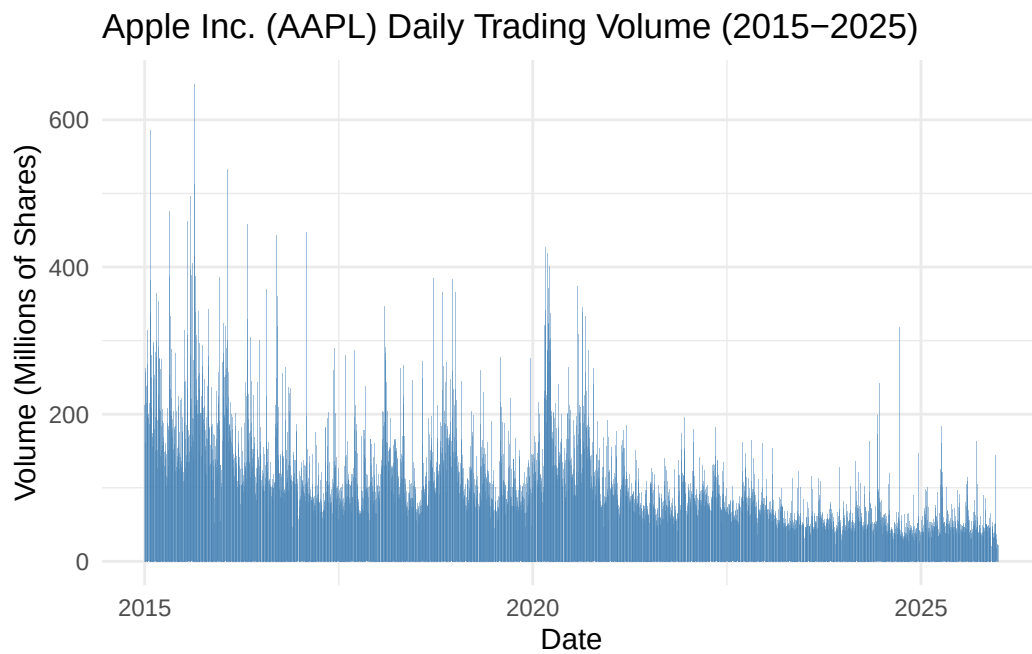
Apple's adjusted closing price grew from approximately \$24 in early 2015 to over \$250 by end of 2025, representing roughly a 10x increase over the decade. Growth was relatively steady from 2015 to 2019, followed by an explosive run during the COVID-era technology boom in 2020-2021. A sharp correction occurred in 2022 before the stock resumed its upward trajectory.

Chart 2 : Annual Returns



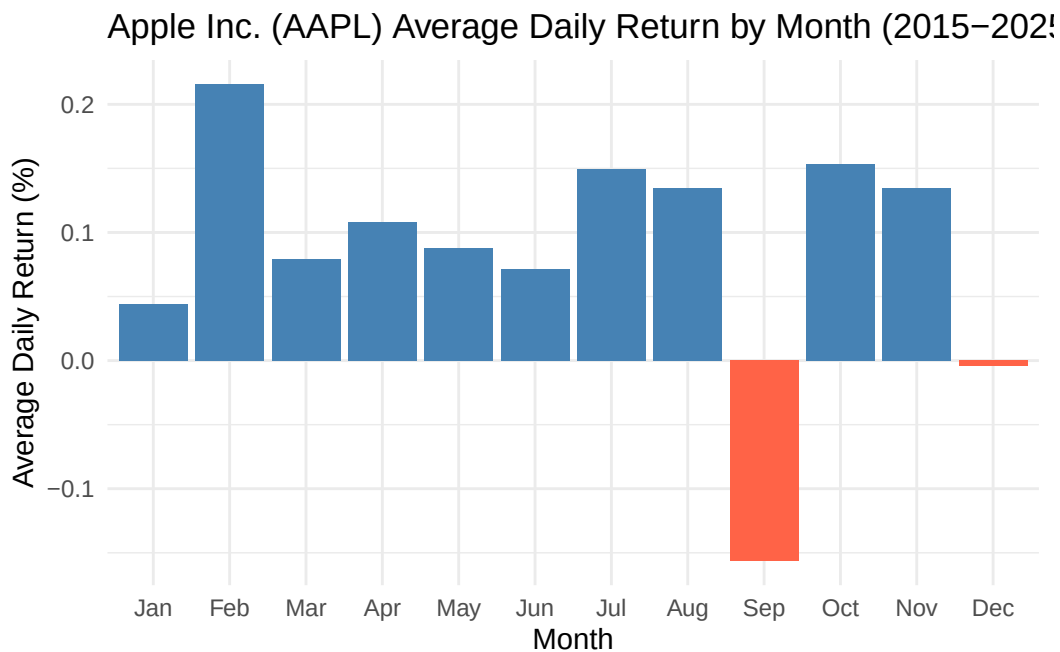
Annual returns varied considerably across the decade. The strongest year was 2020 with approximately 90% growth, driven by surging demand for technology during the pandemic. 2022 was the only significantly negative year at around -27%, reflecting broader market conditions including rising interest rates. Most years delivered positive returns, highlighting Apple's consistent long-term performance.

Chart 3 : Trading Volume Over time



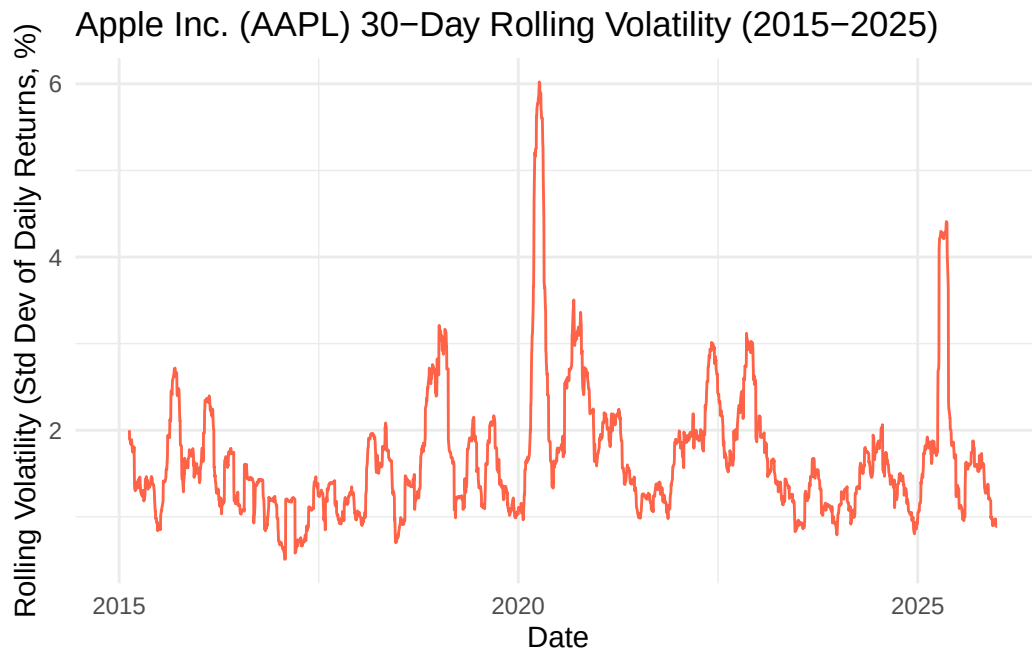
Trading volume was highest in 2015-2016, with several days exceeding 600 million shares, likely reflecting uncertainty around Apple's growth trajectory at the time. Volume spikes are also visible around 2020 during the COVID market volatility. Overall, daily volume has trended lower in recent years, which often indicates a more stable and institutionally held stock with less speculative trading activity.

Chart 4 : Monthly Seasonality



There is a clear seasonal pattern in Apple's daily returns. February is the strongest month on average, while September and October are the only months with negative average daily returns. The poor September performance aligns with the well-documented "September effect" observed broadly in equity markets. Strong performance in the fourth quarter (November-December) may reflect holiday season optimism and iPhone sales cycles.

Chart 5 : Rolling Volatility



The 30-day rolling volatility reveals that Apple’s stock experienced its most turbulent period in early 2020, when the standard deviation of daily returns spiked above 6% as COVID-19 triggered a global market selloff. Volatility elevated again in 2022 during the interest rate hiking cycle. The relatively calm periods of 2017-2019 and 2023-2024 contrast sharply with these spikes, illustrating how external macroeconomic shocks drive short-term risk even in high-quality stocks.

Summary

This exploratory analysis of Apple Inc. stock data from 2015 to 2025 reveals several key insights. Apple delivered exceptional long-term price appreciation of approximately 10x over the decade, though returns varied significantly by year. The COVID-19 pandemic was the defining event of the period, simultaneously driving the highest annual return (2020) and the most extreme volatility spike in the dataset. Seasonal patterns suggest investors should be aware of historically weak September performance. Trading volume has declined over time, consistent with Apple’s maturation into one of the world’s most widely held and stable large-cap stocks. Together, these findings illustrate how both company-specific factors and broader macroeconomic conditions shape stock price behavior over time.